JOHN BOLOGNINO

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RESEARCH INTERESTS

Brain-computer interfaces for communication: ML for decoding language and movement from neural signals (intracortical and EEG), and efficient transformer models for neural time-series data.

EDUCATION

Boston University, Kilachand Honors College **May 2023**
Bachelor of Arts, Computer Science, Psychology Minor GPA: 3.37
Keystone Project (Honors Thesis): "Surveying Machine Learning Techniques for Brain-to-Text Handwriting Communication," conducted with the Dartmouth LISP Lab (see Research Experience)
Relevant Coursework: Algorithms and Data Structures, Machine Learning Principles, Probability, Computation Theory, Natural Language Processing, Algorithmic Data Mining, Deep Learning, Cognitive Neuroscience, Computational Models of the Brain, Neural Models of Memory
Honors: Upsilon Pi Epsilon Computing Honor Society, Dean's List all semesters, Undergraduate Research Award (3x)
Additional Training: Stanford University Artificial Intelligence Professional Program (Non-Degree) June 2026
Stanford Coursework: XCS224N Natural Language Processing with Deep Learning, XCS234 Reinforcement Learning, XCS236 Deep Generative Models

PUBLICATIONS & PRESENTATIONS

- **J. Bolognino**, S. Cohen, E. Bar, et al., "Visible coalitions of neuronal activities in brain-to-text communication via handwriting," in *Proc. SPIE 13517, Seventeenth International Conference on Machine Vision (ICMV 2024)*, 1351718, Feb. 2025. doi:10.1117/12.3056602
- M. Cokbas, **J. Bolognino**, J. Konrad, and P. Ishwar, "FRIDA: Fisheye Re-Identification Dataset with Annotations," in *IEEE International Conference on Advanced Video and Signal-Based Surveillance (AVSS)*, Nov. 2022. arXiv:2210.01582

RESEARCH EXPERIENCE

Learning, Intelligence & Signal Processing Lab, Dartmouth College Remote
Researcher (Part-Time), PI: Peter Chin *January 2022 - November 2024*
In collaboration with Eilon Vaadia, Edmond and Lily Safra Center for Brain Sciences, Hebrew University of Jerusalem

- Designed and led experiments decoding imagined handwriting from motor cortex recordings of a paralyzed participant (Willett et al. data), representing neural activity as raster displays and training CNNs that reached 86% accuracy across 31 characters (96% average on binary tasks).
- Applied Grad-CAM to identify spatiotemporal patterns of neuronal activity driving character classification, supporting the role of dynamic neuronal correlations in encoding characters.
- For Kilachand Honors Keystone thesis, compared neural networks operating on images, image sequences, graphs, and simplicial complexes for character classification; spatiotemporal (image-sequence) models outperformed graph-based models despite the latter capturing more inter-neuron correlations.
- Supervised, with PI Peter Chin, a team of undergraduate researchers from the Hebrew University of Jerusalem in executing experiments.

Visual Information Processing Lab at Boston University Boston, MA
Undergraduate Researcher, PI: Janusz Konrad *June 2021 - January 2022*

- Co-created FRIDA, a published benchmark fisheye re-identification dataset with 10+ citations, licensed by private companies.
- Three-times awarded Undergraduate Research Opportunity Program funding to research adaptation of person re-identification algorithms for use on fisheye images
- Gathered benchmark performance statistics for a novel indoor, multi-camera fisheye dataset for person re-identification on eight state-of-the-art deep learning re-identification algorithms designed using PyTorch, TensorFlow, and FastAI

PROJECTS

- **Electrode-Specific Masking in EEG-to-Text Translation:** November 2025 - Present - Designing a model pre-training methodology leveraging region-specific features of EEG data for EEG-to-Text Translation on the ZuCo dataset (in progress). Modernized the baseline transformer with FlashAttention-2, achieving a 39% increase in training speed with no loss in accuracy. Whitepaper currently in review through Stanford Professional Program.

PROFESSIONAL EXPERIENCE

Savvas Learning Company — Boston, MA
Software Developer, AI-Enabled Learning — *December 2024 - Present*
Associate Software Developer — *June 2023 - December 2024*

- Designed and built a K-12 essay autograding engine (two Gold Stevies, 2026 American Business Awards), including a custom multi-headed transformer architecture (HuggingFace, PyTorch) that consolidated 8 per-task endpoints into 3 and, with an Inf2 migration, cut AWS inference costs by 75%; it now serves 21 classification and regression tasks from 5 endpoints.
- Led model and infrastructure upgrades that reduced average real-time inference latency by ∼50% (800ms to 400ms per batched request), migrating AWS SageMaker endpoints from Inf1 to Inf2 and upgrading base models from BERT-base to DeBERTa-v3.
- Distilled behaviors from frontier LLMs (Claude Haiku and Sonnet) into smaller transformer models that meet production privacy, latency, and explainability requirements, averaging 91% agreement with subject-matter experts across tasks.
- Built an Oral Reading Fluency diagnostic tool using Whisper and Nemotron speech models.
- Deployed and served containerized models via AWS SageMaker real-time endpoints, handling 22 million requests in 2025 for an educational platform used by over 10,000 school districts, with Datadog monitoring and alerting for endpoint health, latency, and model performance.
- Helped implement custom-pretrained PII-scrubbing models that keep model training and evaluation on student data FERPA-compliant.
- Managed versioning, evaluation, and iterative improvement of all production ML models, incorporating feedback from subject-matter expert (teacher) raters into continuous model refinement.
- Mentored the team's summer intern three years running and hold regular office hours mentoring three junior developers.
- Spearheaded and organized monthly paper club for team of 12, discussing recently published academic literature in ML and NLP as they relate to our development objectives; discussions led us to adopt gradient surgery (PCGrad) for multi-task optimization in our scoring model, raising mean per-head accuracy by 4%.

Potoo Solutions — Norwalk, CT
Python Developer/IT Team Member (Part-Time; Full-Time During Breaks) — *May 2019 - June 2023*

- Joined as a high school intern through the Mayor's Youth Employment Program and stayed on through college.
- Developed NLP pipeline using Python libraries NLTK, spaCy, and scikit-learn to assess the likelihood that a product is being sold by an illicit third party using e-commerce data including product reviews.
- Rewrote core ETL processes from Python to Go with asynchronous design to make database update jobs 3× faster and scale to handle a 175% throughput increase as well as bespoke ETL tools for clients such as Adidas and VF. Leveraged AWS Lambda and S3.
- Developed automation to gather e-commerce metrics for over 600 clients with accompanying GUI made with tkinter for account managers to access data unified from 5 vendor APIs and an internal PostgreSQL database in one dashboard, streamlining workflow by over 1,500 hours yearly.

TEACHING

Computer Science Course Assistant — Boston, MA
Department of Computer Science at Boston University — *January 2021 - May 2023*

- Assisted professors in Intro to Computer Science (Spring 2021, Spring 2023) and Probability in Computing courses (Fall 2021 - Fall 2022)
- Facilitated weekly lab practicum and twice-weekly office hours. Graded weekly problem set questions with feedback and responded to students' questions on online course forum
- Helped implement live queue system for Zoom office-hours to adapt to remote learning

Math Tutor, Mathnasium — Stamford, CT — *2016 - 2019*

PROFESSIONAL MEMBERSHIPS

Institute of Electrical and Electronics Engineers (IEEE), Member — *2026 - Present*
Boston Section; Computational Intelligence Society; Signal Processing Society

SKILLS

Programming: Python, Go, Ruby, SQL, JavaScript; **Familiar:** C, R, MATLAB
ML/Data: PyTorch, HuggingFace Transformers, scikit-learn, Whisper, FastAI, CUDA, NumPy, Pandas, Model Distillation, Real-Time Inference
MLOps/Infra: AWS SageMaker, S3, Lambda, EC2, Docker, Datadog, Git
Languages: Spanish (C1), Italian (B2)